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Abstract

This project explores factors that influence salary across various industries. We focused on jobs in tech, management, finance, research, and engineering. The dataset included over 6,000 entries with variables such as age, gender, education level, experience, and job title. We cleaned, standardized, and categorized job titles to simplify analysis. Exploratory analysis revealed consistent salary gaps based on gender and education level. We used regression models—including linear regression, decision trees, random forests, and XGBoost—to predict salary. Among them, XGBoost performed best with the highest accuracy. Experience and job category were the most important predictors. Our visualizations also uncovered the underrepresentation of women in high-paying senior roles. This research highlights inequalities in pay and offers a framework for fairer salary prediction.

Introduction

Gender inequality in Science, Technology, Engineering, and Mathematics (STEM) careers has been a widely discussed issue for many years. While more women are earning STEM degrees and entering technical fields, significant gaps still remain in both representation and pay [piloto_2023]. Men continue to dominate many high-paying roles, especially in areas like computer science, data science, and engineering [fry_kennedy_funk_2021]. At the same time, women are often underrepresented in leadership positions, and even when these women have higher levels of education, they still earn less than men [piloto_2023]. Even worse, even when they are doing similar work, men still receive higher pay. These patterns raise important questions about fairness and opportunity in modern workplaces, especially in industries that are rapidly growing and shaping the future.

It is important to understand whether the wage gap problem is improving over time, or whether some disparities are remaining the same, or even getting worse [kochhar_2023]. These questions are especially relevant for students, educators, and policymakers who are working toward a more inclusive and equitable STEM environment.

This project investigates how age, gender, education level, years of experience, and job category affect salary outcomes. While previous studies have explored some of these variables independently, there is limited analysis that combines them into a unified predictive model, particularly across varied industries and seniority levels. Additionally, most salary analyses remain descriptive in nature, lacking a predictive component that could guide real-world decision-making in HR and policy design. Our goal was to move beyond basic summaries by constructing and comparing machine learning models to predict salary, while also identifying the key features that drive compensation across the workforce.

To move forward with these goals, we used a publicly available dataset from Kaggle, a platform that hosts datasets shared by researchers, companies, and survey projects [kaggle_2024]. The dataset includes the variables job title, gender, salary, education level, and years of experience. This data gave us a detailed view of how men and women are positioned in the workforce, especially in STEM-related fields. It also allowed us to analyze pay patterns and representation across different job roles, skill levels, and locations.

We began our research by cleaning the dataset, which involved standardizing inconsistent string entries, ensuring consistent formatting in job titles, and dropping five rows that contained at least one null value. We also checked for outliers in the salary column using interquartile range (IQR) filtering and found no significant outliers that required removal. Our analysis then followed three structured phases: exploratory data analysis (EDA), feature engineering, and predictive modeling. During the EDA phase, we began by visualizing salary distributions across gender, education levels, and job industries to identify initial patterns and outliers. We observed consistent gender pay gaps and noticeable disparities in compensation by education level and seniority. To quantify these observations, we conducted independent sample T-tests to compare average salaries between male and female professionals, and calculated Cohen's d to assess the effect size of gender on salary. To support our visual findings with statistical rigor, we conducted hypothesis testing using **Welch's t-test** to assess significance and **Cohen's d** to measure effect sizes. These tests allowed us to quantify whether observed disparities (e.g., gender wage gaps at each education level) were statistically significant, and to understand the magnitude of those disparities. These statistical tests reinforced our visual findings and provided empirical support for the presence of inequity.

Next, we moved to feature engineering to prepare the dataset for modeling. We consolidated over 190 unique job titles into broader industry categories (e.g., Tech, Finance, Research) and created seniority groupings (Junior, Mid-level, Senior, Mid-Senior, and Executive) based on job titles to account for career stages. We also binned years of experience into intervals of 5 years, which allowed us to compare salary trends across career phases more consistently. These transformations helped simplify a high-dimensional dataset and made trends easier to interpret while preserving essential variability.

Finally, in the predictive modeling phase, we trained and evaluated five different regression models: Linear Regression, Decision Tree, Random Forest, XGBoost, and Support Vector Machine (SVM). Our aim was to explore a range of modeling approaches that reflect different assumptions about data structure and feature relationships. Linear Regression served as a baseline due to its interpretability and its assumption of linear relationships between predictors and the target variable [southekal_2024]. We included Decision Trees to model non-linear relationships and feature interactions in a rule-based, hierarchical structure that can be visualized and easily interpreted. Random Forest, an ensemble of decision trees, was chosen for its ability to reduce overfitting and improve generalization through bagging and random feature selection [sharma_2020]. XGBoost, a gradient boosting algorithm, was included for its computational efficiency, handling of missing values, and strong performance in high-dimensional data due to sequential error correction [chen_guestrin_2016]. Lastly, we tested a Support Vector Machine (SVM) with a regression kernel to evaluate a distance-based approach, particularly useful when relationships are complex and boundaries between target outputs are not easily defined by linear or tree-based splits [obulesu_mahendra_thrilokreddy_2018]. By using this diverse set of models—spanning linear, tree-based, ensemble, and kernel methods—we aimed to gain a deeper understanding of the mathematical relationships in our data and ensure that our conclusions were not biased by a single modeling framework.

By combining exploratory data analysis, statistical testing, and a range of predictive modeling techniques, this project seeks not only to understand salary disparities but also to provide a scalable framework for fairer, data-driven compensation analysis.Our results showed that gender-based pay gaps still exist in most STEM fields. Men were paid more than women in nearly every category, even when job titles and experience levels were similar. The largest disparities appeared in engineering and data-focused roles, while fields like biology or education-related tech jobs showed smaller gaps. Women were also underrepresented in high-level positions, especially in roles with higher salaries or decision-making responsibilities. However, we also found signs of progress, particularly among younger professionals or those in early-career roles, where the gaps were slightly smaller.

Methods

Our project followed an iterative and structured workflow designed to investigate patterns in salary data and build predictive models grounded in social context. As shown in Figure 1, we began with problem framing and a comprehensive data preparation phase, which involved acquiring, cleaning, and exploring the dataset. This was followed by feature engineering. We then conducted statistical analysis—including t-tests and Cohen's d calculations—alongside data visualizations to uncover disparities by gender, experience, and education. Multiple regression models were trained to predict salary and evaluate the influence of different features. Finally, in the reflection phase, we extended our findings by analyzing CWRU faculty salary data and synthesizing insights into our final presentation. Our process allowed for feedback loops between exploration, modeling, and analysis, ensuring both rigor and interpretability throughout the project.

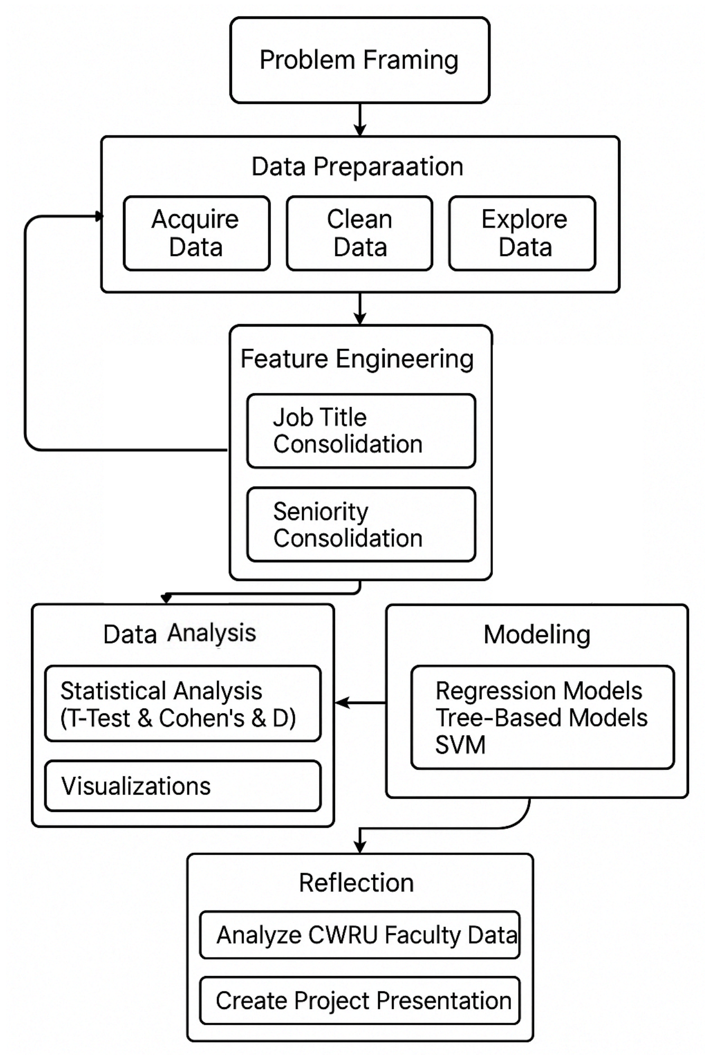

Figure 1. Overall Project Workflow

The dataset used in this project was sourced from Kaggle and contains salary-related data for professionals across various industries and roles [sai_2023]. It includes 6,771 entries with features such as Age, Gender, Education Level, Job Title, Years of Experience, and Salary. The data captures a diverse range of job functions and seniority levels, making it well-suited for analyzing patterns of compensation and inequality. While the dataset does not specify geographic or company-level information, its breadth in individual attributes allowed us to explore key factors influencing salary and build predictive models based on demographic and occupational variables.

After an initial inspection, we identified several inconsistencies and missing values. Specifically, 2–5 missing entries existed across different columns. We removed five rows that contained at least one missing value, resulting in 6,698 remaining rows. We also ensured uniformity in categorical columns: e.g., “PhD” and “phD” were standardized, and gender entries were mapped consistently (e.g., 'm', 'man', 'boy' → 'Male') and then numerically encoded (Male=1, Female=0). To address extreme salary values, we applied Interquartile Range (IQR) filtering.

$$IQR = (Q_3 - Q_1)$$

Here, Q_1 and Q_3 signify the data’s 25th and 75th percentile respectively. The lower bound for filtering is defined as $Q_1 - 1.5 \times IQR$ while the upper bound is defined as $Q_3 + 1.5 \times IQR$. We filtered the data within the bounds $x \epsilon [Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$. The IQR method detected no significant outliers, thus no rows were removed at this step.

Through Exploratory Data Analysis, we explored the distribution of features such as Gender, Education Level, Age, and Job Title**s**. As part of our initial exploration, we visualized the distribution of key features in the dataset to understand trends, identify skewness, and guide further preprocessing decisions. We also generated visualizations such as bar plots, histograms, and box plots to clearly observe disparities and patterns. One major insight during this phase was the high cardinality of the "Job Title" column—there were 193 unique job titles, which posed challenges for direct comparison.

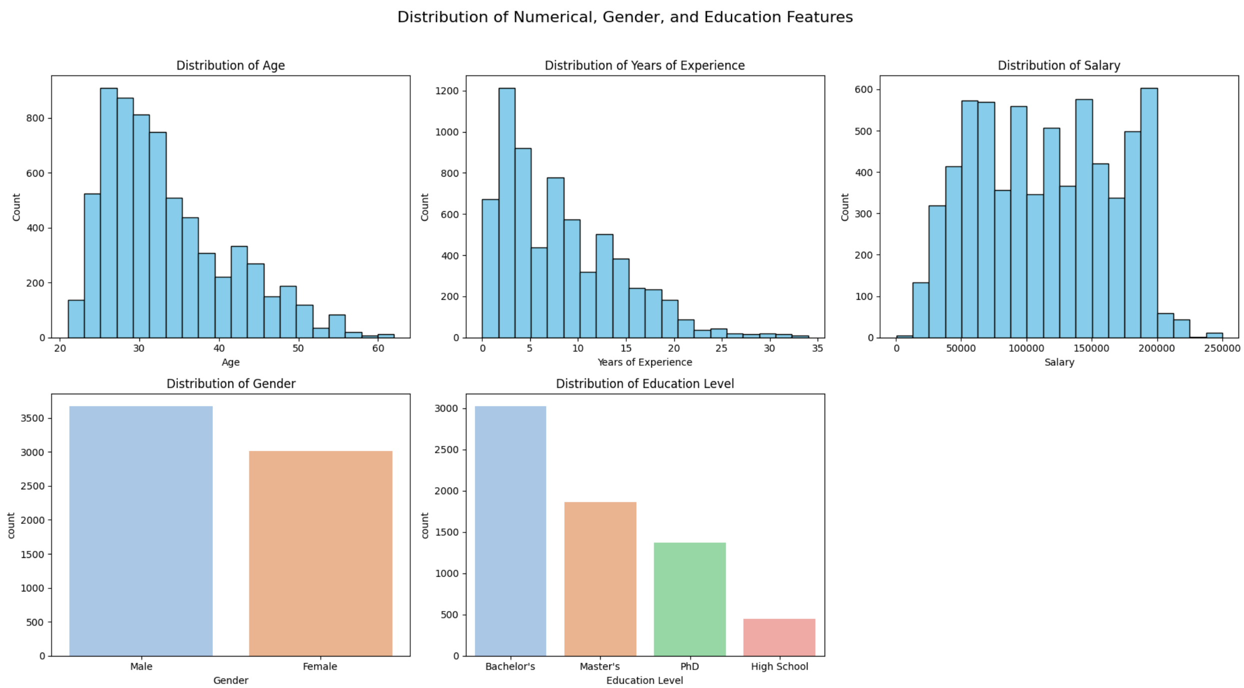

Figure 2. Distributions of Age, Years of Experience, Salary, Gender, and Education Features

Figure 2 shows the distribution of education levels across the dataset. The majority of individuals hold a Bachelor's degree, followed by Master's and PhDs, while only a small proportion have High School as their highest educational attainment. This skew suggests a sample weighted toward higher-educated professionals, consistent with many white-collar industry roles.

Age and Experience distributions are right-skewed, with most individuals falling in their 20s to early 30s and under 10 years of experience. Salary displays a multimodal distribution, with concentrations in the $50k–$100k and $150k–$200k ranges. Gender distribution appears relatively balanced, with a slightly higher number of male entries. These visual patterns helped shape our decisions around binning experience, handling outliers, and stratifying analysis by gender and education level.

From our initial EDA, we engineered several new features to reduce dimensionality and enable more meaningful group comparisons. First, we consolidated 193 unique job titles into broader **industry categories**—such as Tech, Finance, Management, Research, and Engineering—based on keyword patterns and external taxonomies. We also extracted **seniority levels** (e.g., Junior, Mid, Senior, Executive) using rule-based mapping from job title strings. To simplify analysis of experience, we binned **Years of Experience** into five groups: 0–5, 6–10, 11–15, 16–20, and 21+ years.

Categorical variables were standardized and numerically encoded. Gender was encoded as binary, and Education Level was converted to an ordinal scale (High School = 0 to PhD = 3). These transformations allowed us to make statistically sound comparisons across groups and prepared the dataset for downstream analysis and modeling.

With these engineered features, we structured our analysis into two main comparisons: one examining **Education Level and Years of Experience**, and the other focusing on **Industry Category and Seniority Level**. This split enabled more targeted insights into how educational background and career progression influence salary across demographic groups.

To support our visual findings with statistical rigor, we conducted hypothesis testing using **Welch's t-test** to assess significance and **Cohen's d** to measure effect sizes.

We initially considered two strategies to manage the high variability in job titles: (1) focusing only on the five most frequent roles, or (2) consolidating titles into broader industry groups. We ultimately pursued the second strategy, which enabled more robust modeling and analysis while retaining a meaningful diversity of roles.

Following feature engineering and analysis, we proceeded to build predictive models to estimate salary based on the cleaned dataset. A mix of models was chosen to assess how well different algorithmic paradigms—ranging from linear (Linear Regression), to ensemble-based (Random Forest, XGBoost), to margin-based (SVM)—capture the complex patterns in salary data. Each model was evaluated using two key metrics on the test set:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

The Root Mean Squared Error (RMSE) provides a measure of average prediction error in the same units as the target (salary), with a greater penalty for large errors—making it useful for understanding real-world prediction accuracy. The Coefficient of Determination R^2 indicates the proportion of variance in salary explained by the model, with values closer to 1 representing stronger explanatory power. Together, these metrics offer a comprehensive evaluation of model performance [chugh_2020].

After completing our main analysis, we extended the project by applying the same job classification and analysis framework to a publicly available CWRU faculty salary dataset. This provided a point of comparison to real-world academic salaries and offered an opportunity to reflect on the generalizability of our approach. We also used our insights to create a polished presentation for sharing findings with peers and stakeholders.

Timeline of Progress

- Late March:
  - Found relevant data to accompany the dataset.
  - Began basic data pre-processing and cleaning
- Early April:
  - Began drafting the code to form figures relevant to the data
  - After cleaning, observed values to be kept, dropped, and how to optimize the data while keeping it clean.
- Late April:
  - Finalized code and generated relevant figures
  - Drew conclusions from the observations from the graphs
  - Presented findings.

Results

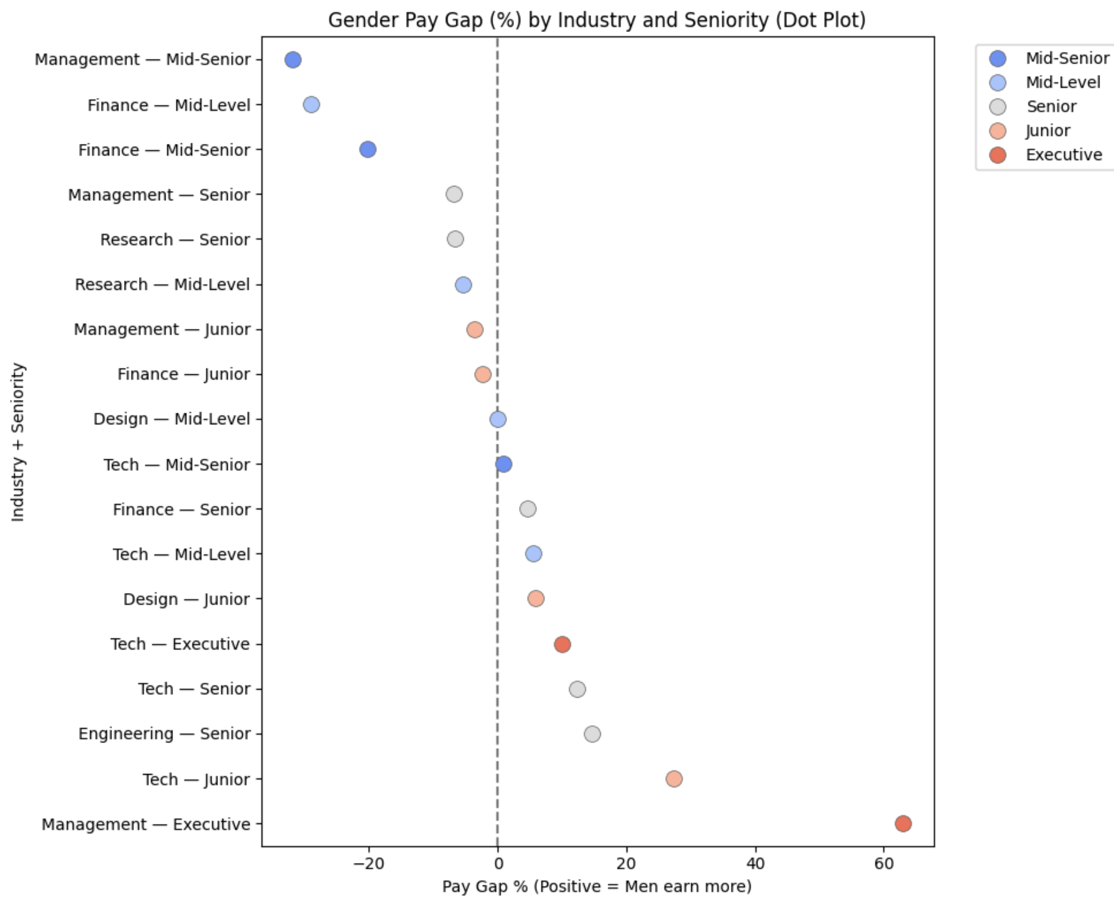

Figure 3. Dot Plot of Gender Pay Gap (%) by Industry and Seniority

Figure 3 presents a dot plot of gender pay gaps (in percentage) across various industry and seniority combinations. Positive values indicate that men earn more than women. The dot plot illustrates that gender pay gaps vary widely across industry and seniority levels. In Design, only the Junior level shows a gap (~6%) where women earn less; other levels show no significant differences. Engineering includes only the Senior level, revealing a 14% gap favoring men. In Finance, the Mid-Level role displays the most substantial reverse gap (~30%) in favor of women, while the Junior level also shows a slight reverse gap. Senior roles appear balanced, whereas Executive positions show a ~20% gap favoring men. Management presents mixed results, with Mid-Level showing an unusually high male advantage (~380%), potentially due to outliers; meanwhile, Executive roles display a ~60% gap in favor of women, and other levels remain relatively balanced or slightly favor women. In Research, both Mid-Level and Senior levels show reverse gaps of ~5–6%, where women earn slightly more than men. The Tech sector demonstrates consistent male-favoring gaps at all seniority levels: ~27% at the Junior level and between 1–10% at Mid-Level, Senior, and Executive roles.

Overall, most roles exhibit positive pay gaps (indicating men earn more), with notable exceptions in mid-level Finance, mid-senior Management, and Research. The Junior level shows varied patterns across industries, but the pronounced gap in Tech — Junior suggests early career disparities. The plot also highlights how the pay gap tends to widen with seniority in certain fields, particularly in Tech and Management.

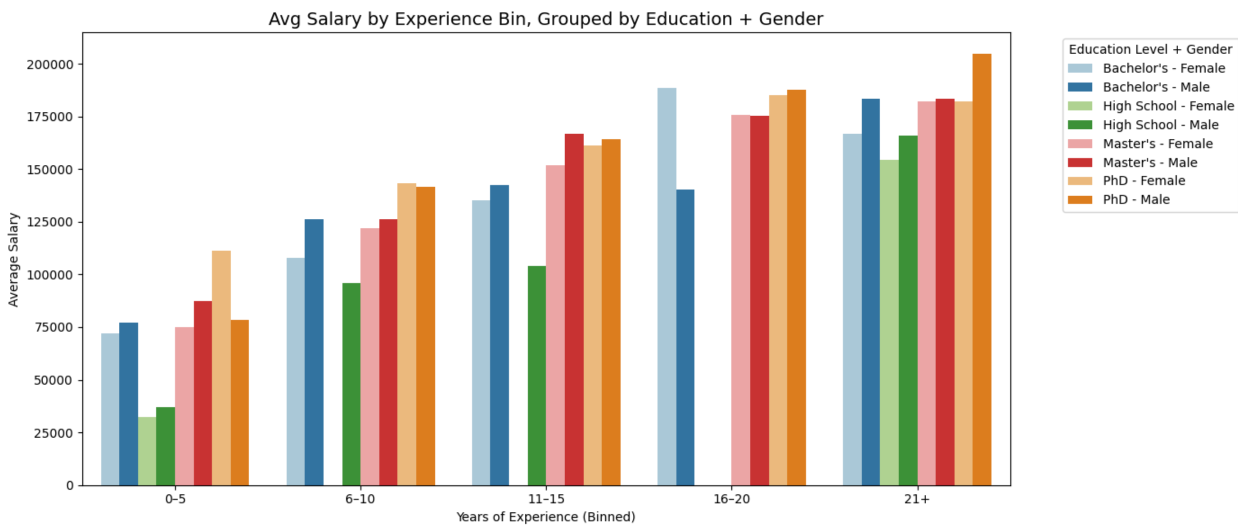

Figure 4. Average Salary by Experience Bin, Grouped by Education & Gender

The gender wage gap is present across all education levels, but its magnitude and direction vary depending on years of experience. It is a more in-depth version of the first bar chart in the results section. It shows the difference in salary between women and men with the same degree and years of experience.

For individuals with 0–5 years of experience, the gap is relatively small. PhD-holding females earn more than their male counterparts in this group, while males with a Bachelor's and High School education slightly out-earn females.

As experience increases to 6–10 and 11–15 years, the gap becomes more pronounced, particularly among those with a Bachelor's or High School education, where it largely favors males. In contrast, Master's and PhD holders continue to show either minimal gaps or even instances where females earn more.

By the 16–20 year mark, the gender gap widens in the Bachelor's group, with males earning significantly more than females. However, in the Master's and PhD categories, salaries remain relatively balanced, though they start to slightly favor males.

At 21+ years of experience, the gap is most evident. PhD males earn the highest salaries overall, while females across all education levels earn less. Nevertheless, women with Master's and PhD degrees remain the closest to closing the gap. This trend suggests that the gender wage gap tends to widen with seniority, especially in lower education groups.

As we observe the graph, it shows that experience and education both positively influence salary, but education has a compounding effect over time. The gap between education levels widens with experience, especially noticeable between high school and PhD holders. Investing in higher education generally leads to higher long-term earnings, with PhDs having the greatest return as experience increases.

We analyzed wage gaps between males and females across different education levels and years of experience. For individuals with only a high school education, the gender wage gap is generally positive, indicating that males tend to earn more. However, the data in this category is limited due to a high number of missing values (NaNs). The largest gap in this group is observed among those with over 21 years of experience, where males earn $11,712 more than females—a 7.59% difference.

Among those with a Bachelor's degree, gender wage gaps remain consistently positive, again favoring males. The most pronounced gap appears at the 21+ years of experience mark, reaching $16,750 (10.05%). Interestingly, this trend reverses in the 16–20 years experience group, where females earn significantly more than males, with a negative gap of $47,987 (−25.47%).

For Master's degree holders, the gender pay gap also generally favors males, although the percentage differences are smaller compared to other education levels. Notably, the gap is nearly zero for those with 16–20 years of experience, and slightly reverses in the 6–10 year range, with females earning marginally more.

At the PhD level, results are mixed. In the 0–5 year experience group, females actually earn more than their male counterparts, reflected by a −29.50% gap. However, for those with over 21 years of experience, the gap shifts in favor of males, who earn $22,313 more, a 12.24% difference.

Moving from descriptive and statistical analysis to predictive modeling, we trained and evaluated a variety of regression models to forecast salary based on demographic and professional features. Table 1 illustrates the performance of these models.

INSERT TABLE 2 (EMAIL

Discussion

Our analysis revealed persistent and compounding salary inequalities across gender, experience, education, and job industry. While our visualizations and statistical models confirmed these patterns, interpreting their significance reveals deeper insights about structural inequities in the workplace. One of the most prominent findings, as shown in all figures, is that women consistently earn less than men even when controlling for education and experience. Despite holding similar degrees and years of experience, female professionals were underrepresented in senior roles and high-paying categories, particularly in fields like Technology and Finance. This disparity was visualized through our double bar graphs comparing average salaries by gender and education level, which clearly showed a salary gap at every level of education. These trends suggest that credential attainment alone does not level the playing field, reinforcing arguments made in broader literature about invisible barriers and systemic bias in the workforce.

Another significant pattern emerged when we examined the relationship between experience and salary. Our salary vs. experience line plots, disaggregated by gender, showed that men experience steeper salary growth over time. Even when the initial gap may appear small, the cumulative effect over several decades results in substantial income divergence. This aligns with the concept of compounding disparities, where even modest early-career differences in raises or promotions can translate to major lifetime earnings gaps. These patterns raise important questions about internal practices such as promotion cycles, performance evaluations, and raise structures that may favor certain demographics while marginalizing others. While our dataset did not include these organizational-level variables, their absence itself highlights an area for deeper inquiry.

Our strategy to consolidate 193 unique job titles into broader industry categories (Tech, Management, Finance, Engineering, Research, and Executive) allowed us to overcome the challenge of high-cardinality categorical variables and reveal higher-level trends. Figure 5 illustrates that Tech and Finance consistently had the highest average salaries, but also the largest gender-based wage disparities. Education level and age followed, while gender had relatively lower predictive importance in the model. However, this does not mean gender isn't a significant factor—instead, it may operate more subtly through mechanisms like access to roles and promotion tracks, rather than direct salary assignments.

That said, we recognize limitations in our approach. The job title consolidation was partly subjective and could benefit from a standardized occupational taxonomy. We also lacked contextual variables like job location, part-time vs. full-time status, company size, and performance metrics, which may account for some of the salary variance. Nonetheless, our results aligned with documented real-world inequalities and offered replicable, data-driven evidence for examining pay equity.

Moving on, the comparison of regression models, as seen on Table 1, reveals notable differences in predictive performance. Ensemble methods like XGBoost and Random Forest outperformed all other models, achieving the lowest RMSE values (18,330 and 18,622, respectively) and the highest R^2 scores (0.869 and 0.865). These tree-based models are inherently better at capturing complex, nonlinear relationships and handling feature interactions—advantages well-suited to the diverse and structured features in our dataset, such as categorical variables (Education Level, Industry Category, Seniority) and binned experience. Their ability to model these interactions without extensive preprocessing likely contributed to their superior performance.

In contrast, Linear Regression, Ridge, and Lasso Regression all clustered around an R^2 of 0.68, with RMSE values exceeding 28,600. These models assume a linear relationship between predictors and the outcome variable, which may be an oversimplification given the hierarchical and categorical nature of many inputs. Although regularization in Ridge and Lasso prevents overfitting, they offer limited benefit when the true relationship is highly nonlinear. Finally, the Support Vector Machine (SVM) model performed significantly worse ($R^2 = 0.019$), suggesting that it was poorly suited to the scale, dimensionality, or structure of our features. This may be due to the SVM's sensitivity to feature scaling or the absence of kernel tuning.

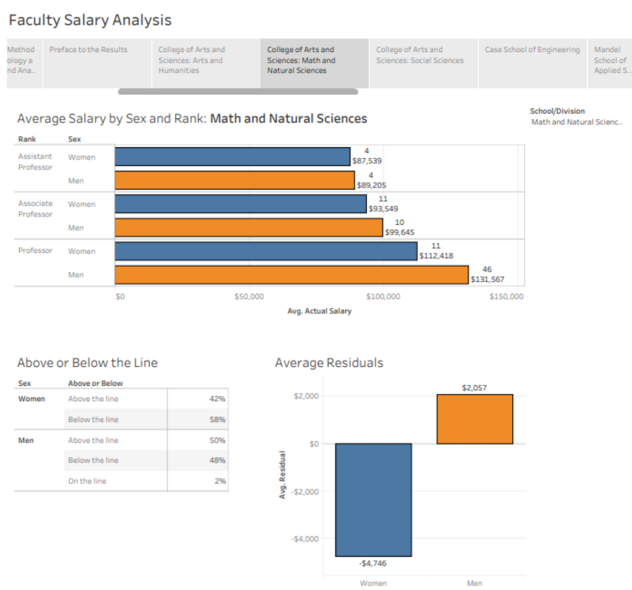

Figure 5. Dashboard of CWRU Faculty Salary Analysis

58% of women are below the average salary 48% of men are below the average salary. This shows how the gender gaps are affecting the real world. As demonstrated by the residual plot for men, the average residual is +$2,057, while for women, the average residual is –$4,746. Suggesting that men are earning more than what the predictive model is estimating. While women are earning less than what the predictive model is estimating. The bar graph further highlights the point that gender gaps increase with years of experience as the gap between women and men for the role of Professor is the biggest compared to the other levels of positions at Case Western Reserve University.

Conclusion

This project revealed persistent salary disparities shaped by gender, education level, and work experience across a wide range of industries. While higher degrees and longer tenure generally correlated with increased earnings, these benefits were not distributed evenly, particularly among women, who continued to earn less even with comparable qualifications. The compounding nature of experience over time further widened income gaps, pointing to structural factors beyond individual merit. Some industries exhibited especially stark inequalities, highlighting the need for sector-specific scrutiny. Through targeted visualizations and statistical analysis, we were able to surface patterns of inequity that might otherwise remain obscured.

Looking ahead, future work could enrich this framework by including additional variables such as job location, performance ratings, or organizational size. Incorporating longitudinal data would allow for tracking wage trajectories over time, while algorithmic fairness tools could help identify and mitigate predictive bias. Ultimately, this project reinforces the importance of grounding compensation practices in data-driven accountability and demonstrates how thoughtful, transparent analysis can support more equitable workforce decisions.

Roles

Ninjin Bilgee

Ninjin contributed to all stages of the project, particularly in the ideation, EDA, and preprocessing phases. She was primarily responsible for designing the final report and presentation slides, ensuring our findings were clearly communicated. She also helped standardize job titles and improve clarity in visual storytelling.

Laura Chen

Laura was actively involved in the EDA and preprocessing phase and contributed significantly to the contextual research, particularly by exploring CWRU faculty and staff salary data as a possible extension of our analysis. Her efforts helped bridge the relevance of our findings to local institutional settings.

Zelan Eroz Espanto
Zelan contributed across all phases, with a strong focus on machine learning model development, predictive analysis, and data visualizations. She implemented and evaluated the regression models (Linear, Decision Tree, Random Forest, XGBoost) and led the creation of visuals that linked salary to experience, education, gender, and industry category.

Sravya Mandava
Sravya initiated the project idea and played a central role in coordinating the team's efforts throughout the timeline. She helped with data cleaning and ideation, synthesized insights across team members, and ensured that the final narrative and deliverables were cohesive and well-aligned with our goals.

References

Provide a list of the references you used to support your points throughout the report. Do not cite Wikipedia (if you used Wikipedia, find the resource cited there, make sure that it is indeed supporting your point, then cite that resource). Literature is not always to be used because they have statements that support your points, you can also be critical of the literature and/or use it to show that the state-of-the-art is not strong.

Instructor's Additional Notes

1. Your report should not be any longer than 12 pages, excluding the list of references. It is ok if you need a page or 2 if you have some exceptional novel figures you are adding.
2. Your report should include at most 5 figures and tables in total. You can combine multiple figures into a single figure by using separate panels, but each figure should be coherent within itself. You must have figure captions and if you have multiple figures then you need to describe each of them and have labels for each.
3. Figure captions should have a title (in bold) followed by a clear description of how to read the figure. The purpose of the caption is to help the reader understand the figure. Your observations from the figure should not be included in the caption, they should be stated in the text.
4. Figures should be clearly visible and all material displayed in the figure should be useful to convey an idea and or to make an observation. You should not include unreadable or irrelevant figures just because they look fancy. You should include the maximum/minimum number of Figures to tell your story (no more or less).

5. You should definitely mention your figure or table in the text. It could be what it conveys or what it shows/results.
6. Label your code chunks. You can use R, Python, or both. Your code should be integrated in this report. In the final report, I don't want to see the code in the final output, but you need to turn in the .qmd file so I can see your code. It must be commented on well.
7. Use a .bib file for references. You can change the .qmd file. Check out the Quatro website for ways to improve output. You can change output to html if need be.